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main.py

```python
import streamlit as st
import numpy as np
import matplotlib.pyplot as plt

st.markdown("""
<style>

/* Main background */
.stApp {
    background: linear-gradient(135deg, #0f2027, #203a43, #2c5364);
    color: white;
}

/* Title */
h1 {
    text-align: center;
    color: #00ffd5;
}

/* Cards */
.css-1d391kg {
    background-color: rgba(255, 255, 255, 0.05);
    border-radius: 15px;
    padding: 10px;
}

/* Buttons */
.stButton>button {
    background-color: #00ffd5;
    color: black;
    border-radius: 10px;
    height: 3em;
    width: 100%;
}

/* Metrics */
[data-testid="stMetric"] {
```

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main.py

main.py > ...

```
[data-testid="stMetric"] {
    background-color: rgba(255,255,255,0.05);
    padding: 10px;
    border-radius: 10px;
}

</style>
""", unsafe_allow_html=True)

# ----------------------------
# functions
# ----------------------------
def calc_gibbs(h, s, t):
    return h - t * s

def check_result(g):
    if g < 0:
        return "Spontaneous"
    elif g > 0:
        return "Not Spontaneous"
    else:
        return "Equilibrium"

def condition_analysis(h, s):
    if h < 0 and s > 0:
        return "Always spontaneous"
    elif h > 0 and s < 0:
        return "Never spontaneous"
    elif h < 0 and s < 0:
        return "Spontaneous at low temperature"
    elif h > 0 and s > 0:
        return "Spontaneous at high temperature"
    else:
        return "Special case"

# ----------------------------
```

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main.py

main.py > ...

```python
# ----------------------------
# page setup
# ----------------------------
st.markdown("<h1> Gibbs Free Energy Analyzer</h1>", unsafe_allow_html=True)
st.markdown("<p style='text-align:center;'>Advanced Thermodynamic Visualization Tool</p>", unsafe_allow_html=True)

# ----------------------------
# sidebar inputs
# ----------------------------
st.sidebar.header("Input Parameters")

h = st.sidebar.number_input("ΔH (Enthalpy)", value=50.0)
s = st.sidebar.number_input("ΔS (Entropy)", value=0.2)
t = st.sidebar.slider("Temperature (K)", 200, 1000, 300)

run = st.sidebar.button("Run Analysis")

# ----------------------------
# main content
# ----------------------------
if run:

    # ========================
    # RESULT SECTION
    # ========================
    st.markdown("## Analysis Section")

    g = calc_gibbs(h, s, t)

    col1, col2, col3 = st.columns(3)

    col1.metric("ΔG Value", round(g, 2))
    col2.metric("Temperature (K)", t)
    col3.metric("Condition", "See below")

    st.markdown("### 📘 Detailed Condition")
    st.info(condition_analysis(h, s))
```

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main.py

main.py > ...

```python
    st.markdown("### 📋 Detailed Condition")
    st.info(condition_analysis(h, s))

    result = check_result(g)

    if result == "Spontaneous":
        st.success("Reaction is Spontaneous (ΔG < 0)")
    elif result == "Not Spontaneous":
        st.error("Reaction is Not Spontaneous (ΔG > 0)")
    else:
        st.info("System is at Equilibrium")

    st.markdown("---")

    # ========================
    # GRAPH SECTION
    # ========================
    temps = list(range(200, 1000, 10))
    values = [calc_gibbs(h, s, temp) for temp in temps]

    fig, ax = plt.subplots()

    ax.plot(temps, values, linewidth=2)
    ax.axhline(0, linestyle="--")

    ax.fill_between(temps, values, 0, where=[v < 0 for v in values], alpha=0.3)
    ax.fill_between(temps, values, 0, where=[v > 0 for v in values], alpha=0.3)

    ax.scatter(t, g, s=100)

    ax.set_facecolor("#111111")
    fig.patch.set_facecolor("#111111")

    ax.set_xlabel("Temperature (K)")
    ax.set_ylabel("ΔG")
    ax.set_title("ΔG vs Temperature")

```

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main.py

main.py > ...

```python
    ax.set_title("ΔG vs Temperature")

    st.pyplot(fig)

    # ========================
    # EQUILIBRIUM
    # ========================
    st.markdown("## Equilibrium Analysis")

    eq_found = False

    for i in range(1, len(values)):
        if values[i-1] * values[i] < 0:
            st.info(f"Equilibrium occurs near {temps[i]} K")
            eq_found = True
            break

    if not eq_found:
        st.warning("No equilibrium point in this range")

    st.markdown("---")

    # ========================
    # SMART INSIGHT
    # ========================
    st.markdown("## Smart Insight")

    if s != 0:
        eq_temp = h / s

        if eq_temp > 0:
            st.info(f"Equilibrium Temperature ≈ {round(eq_temp, 2)} K")

            if t < eq_temp:
                st.write("Currently below equilibrium → behavior may change at higher temperature.")
            else:
                st.write("Currently above equilibrium → reaction behavior already shifted.")
```

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main.py

main.py > ...

```python
            else:
                st.write("Currently above equilibrium → reaction behavior already shifted.")

    st.markdown("---")

    # ========================
    # PHASE MAP (WOW PART)
    # ========================
    st.markdown("## Spontaneity Phase Map")

    temp_range = np.linspace(200, 1000, 60)
    entropy_range = np.linspace(-1, 1, 60)

    G = []

    for s_val in entropy_range:
        row = []
        for t_val in temp_range:
            row.append(calc_gibbs(h, s_val, t_val))
        G.append(row)

    G = np.array(G)

    fig2, ax2 = plt.subplots()

    heatmap = ax2.imshow(
        G,
        extent=[200, 1000, -1, 1],
        aspect='auto'
    )

    ax2.set_xlabel("Temperature (K)")
    ax2.set_ylabel("Entropy (ΔS)")
    ax2.set_title("Spontaneity Regions")

    plt.colorbar(heatmap, label="ΔG")

    st.pyplot(fig2)
```

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main.py

```python
        extent=[200, 1000, -1, 1],
        aspect='auto'
    )

    ax2.set_xlabel("Temperature (K)")
    ax2.set_ylabel("Entropy (ΔS)")
    ax2.set_title("Spontaneity Regions")

    plt.colorbar(heatmap, label="ΔG")

    st.pyplot(fig2)

    st.markdown("---")

    # ========================
    # INTERPRETATION
    # ========================
    st.markdown("## Interpretation")

    st.write("• Negative ΔG → Spontaneous region")
    st.write("• Positive ΔG → Non-spontaneous region")
    st.write("• The boundary between them represents equilibrium")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\anshi\OneDrive\Desktop\streamlit_project> streamlit run main.py

  You can now view your Streamlit app in your browser.

  Local URL: http://localhost:8501
  Network URL: http://172.25.216.141:8501
```

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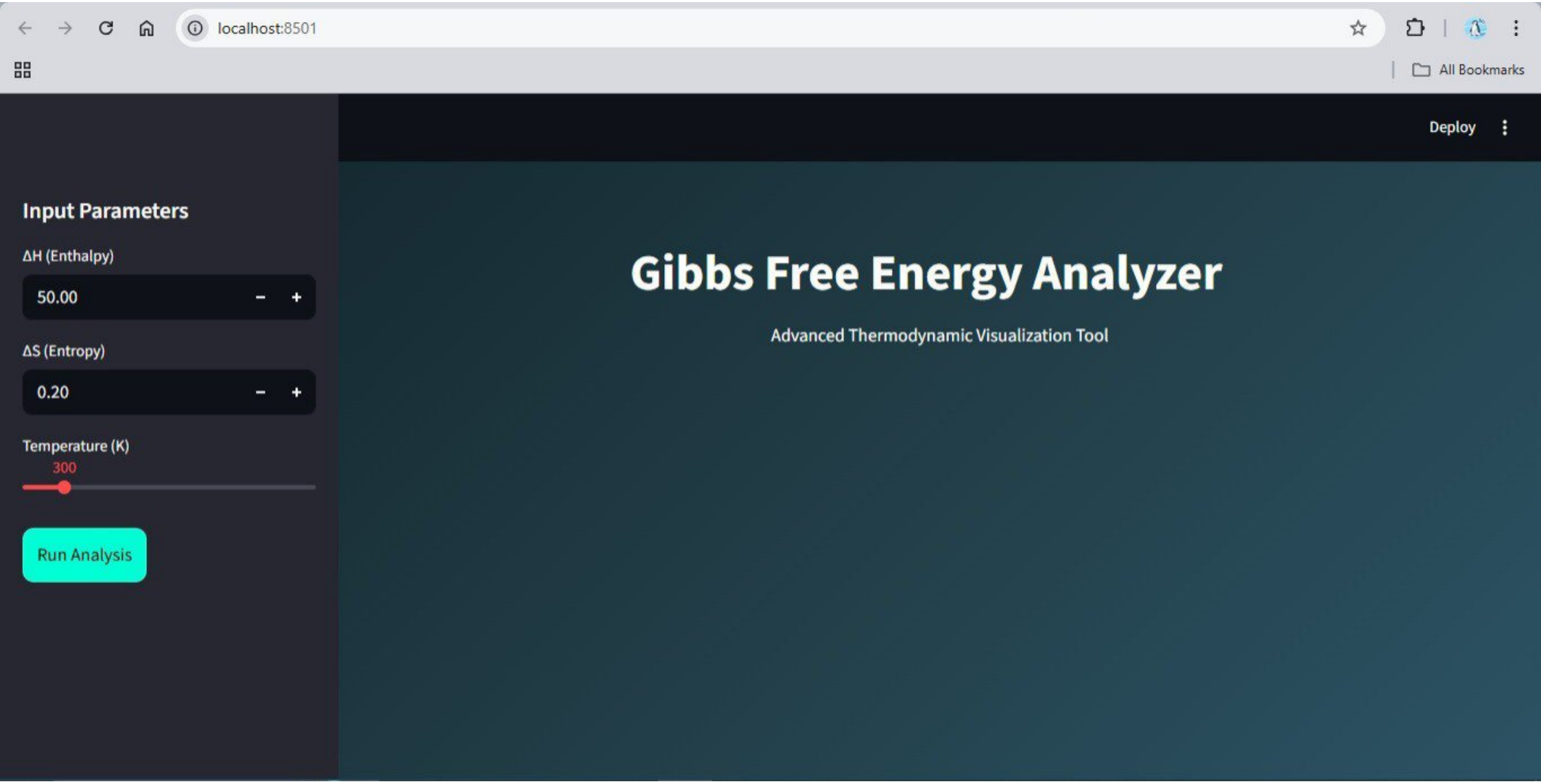
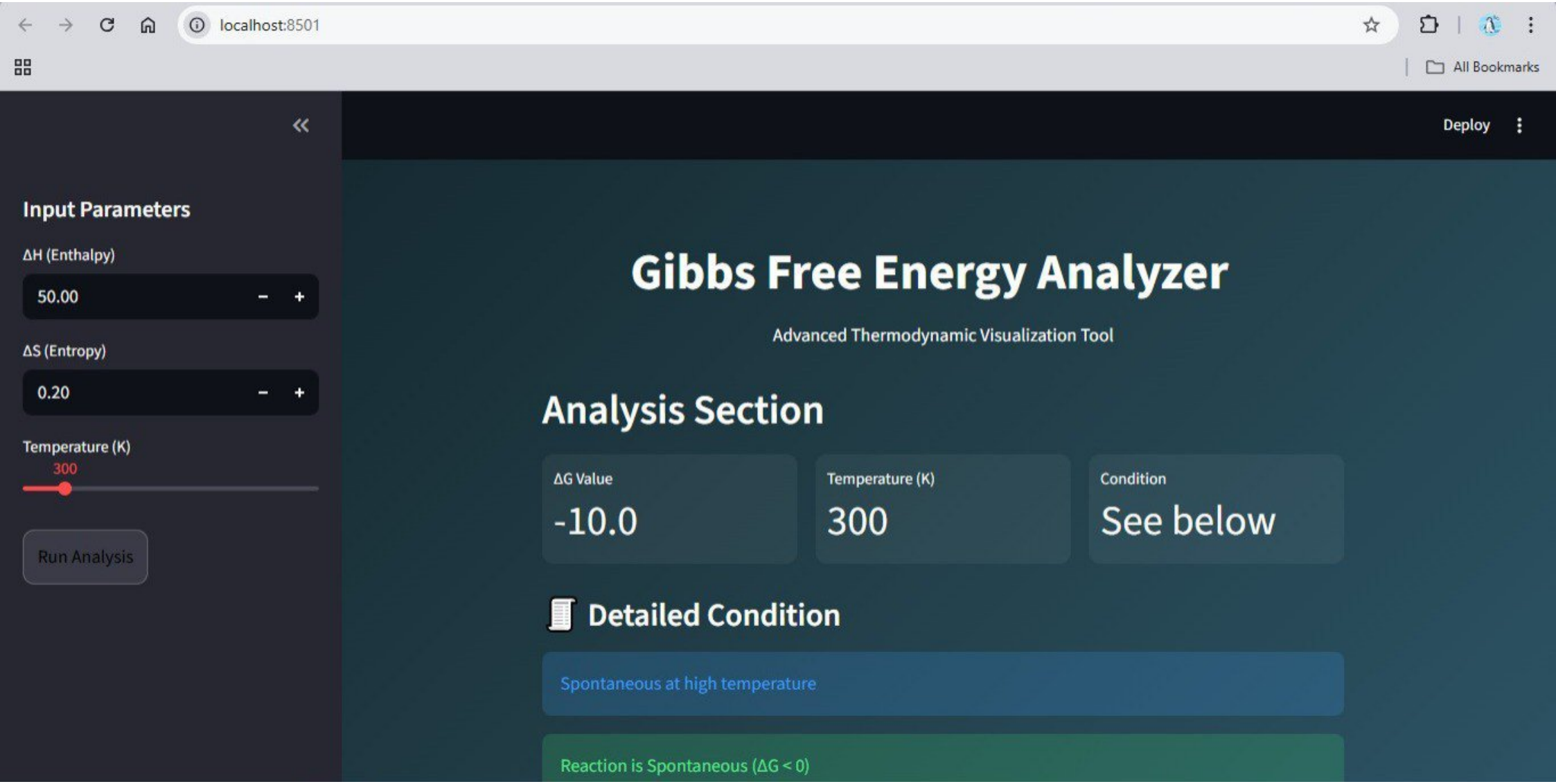
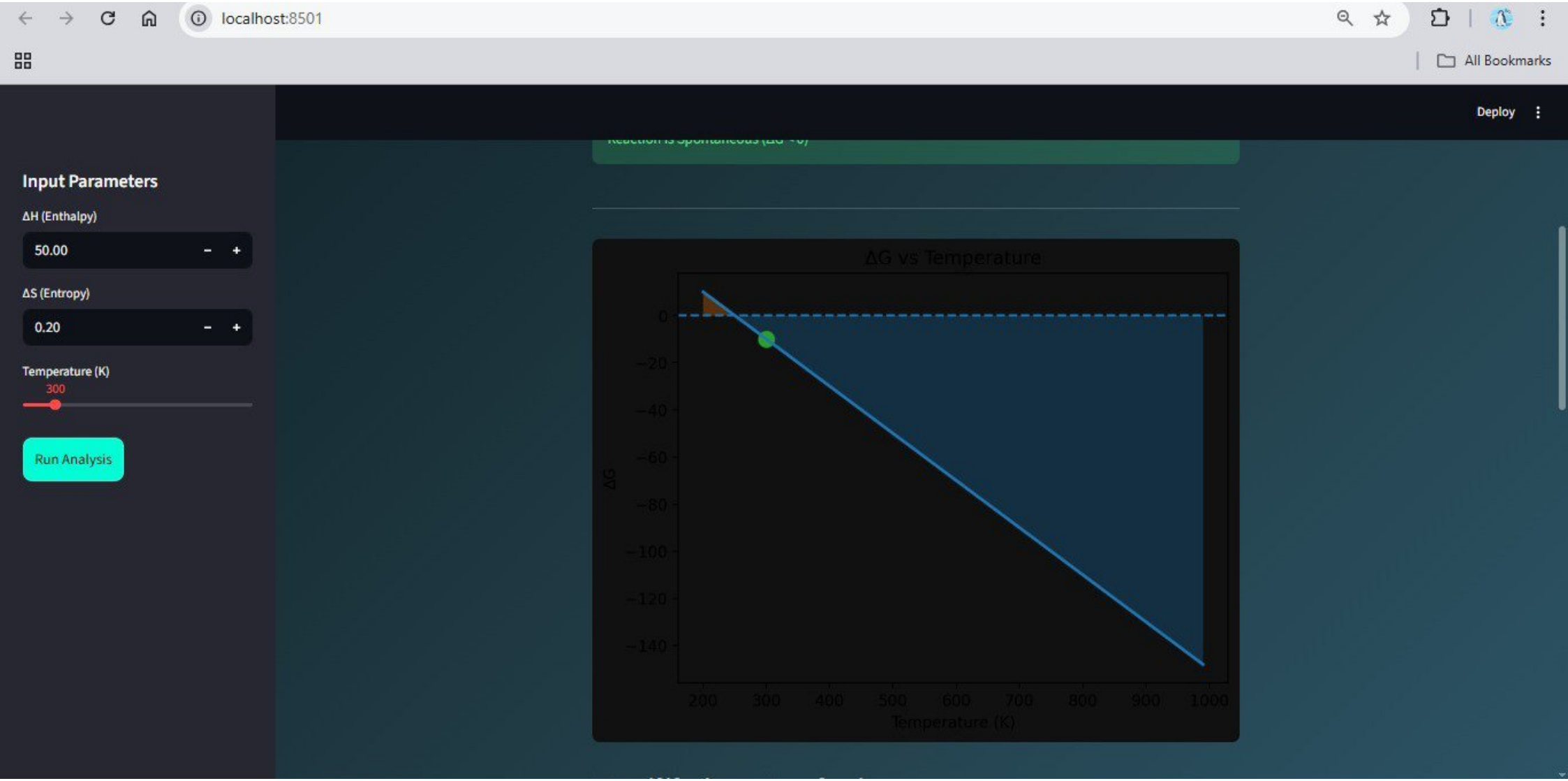

localhost:8501
All Bookmarks
Deploy
Input Parameters
ΔH (Enthalpy)
50.00
ΔS (Entropy)
0.20
Temperature (K)
300
Run Analysis
Gibbs Free Energy Analyzer
Advanced Thermodynamic Visualization Tool


localhost:8501
All Bookmarks
Deploy
Input Parameters
ΔH (Enthalpy)
50.00
ΔS (Entropy)
0.20
Temperature (K)
300
Run Analysis
Gibbs Free Energy Analyzer
Advanced Thermodynamic Visualization Tool
Analysis Section
ΔG Value
-10.0
Temperature (K)
300
Condition
See below
Detailed Condition
Spontaneous at high temperature
Reaction is Spontaneous (ΔG < 0)


localhost:8501
All Bookmarks
Deploy
Input Parameters
ΔH (Enthalpy)
50.00
ΔS (Entropy)
0.20
Temperature (K)
300
Run Analysis
ΔG vs Temperature
0
−20
−40
−60
−80
−100
−120
−140
ΔG
200
300
400
500
600
700
800
900
1000
Temperature (K)

localhost:8501

All Bookmarks

Deploy

Input Parameters

ΔH (Enthalpy)

50.00

ΔS (Entropy)

0.20

Temperature (K)

300

Run Analysis

Equilibrium Analysis

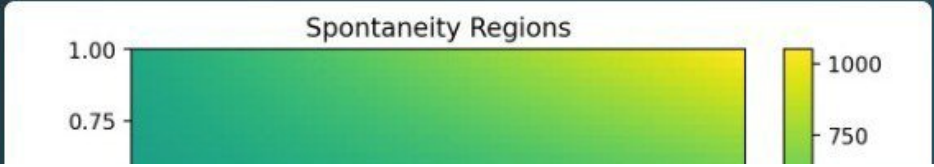
No equilibrium point in this range

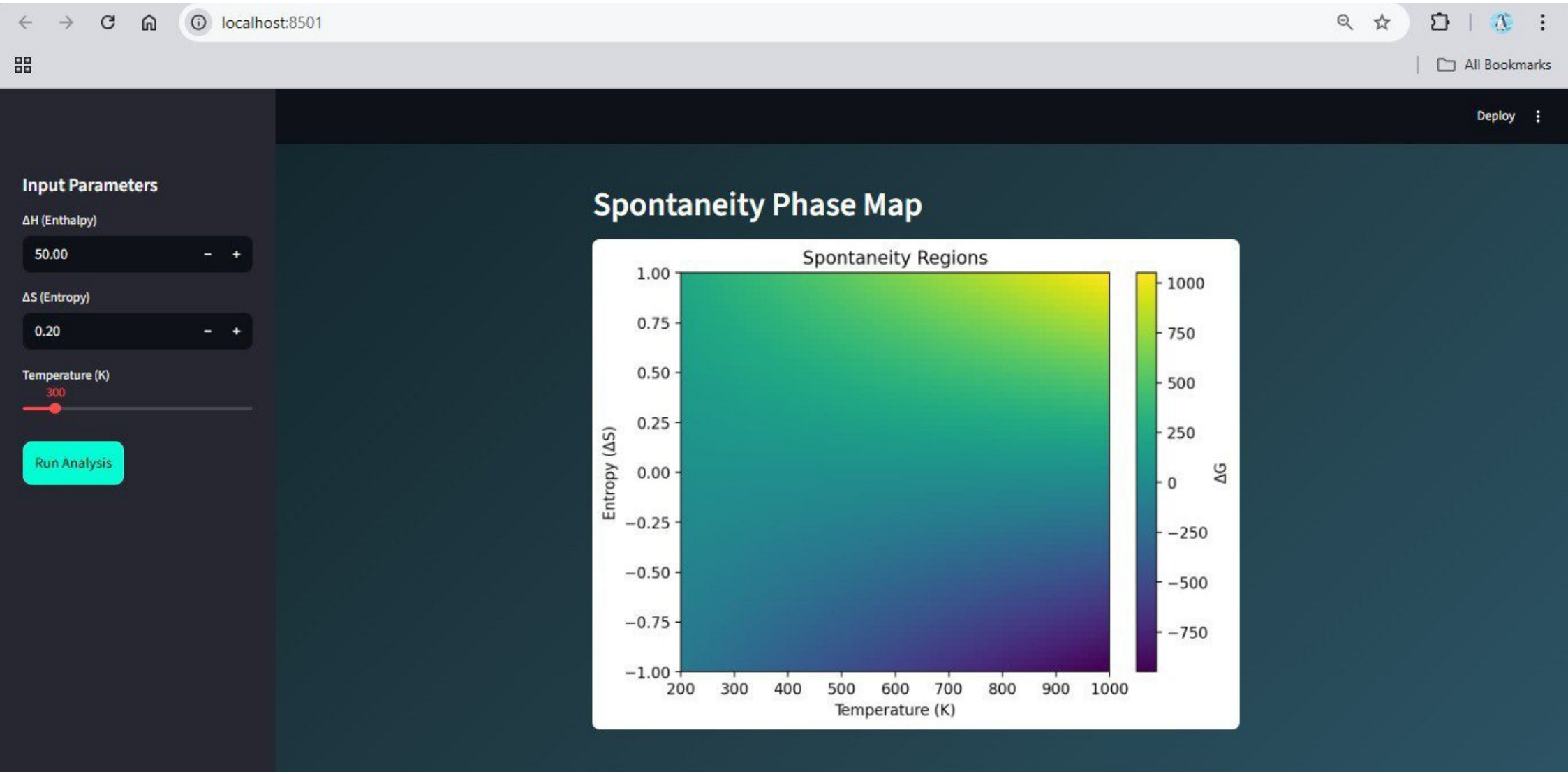
Smart Insight

Equilibrium Temperature ≈ 250.0 K

Currently above equilibrium → reaction behavior already shifted.

Spontaneity Phase Map




localhost:8501
All Bookmarks
Deploy
Input Parameters
ΔH (Enthalpy)
50.00
ΔS (Entropy)
0.20
Temperature (K)
300
Run Analysis
Spontaneity Phase Map
Spontaneity Regions
Entropy (ΔS)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
200
300
400
500
600
700
800
900
1000
Temperature (K)
1000
750
500
250
0
−250
−500
−750
ΔG

localhost:8501

All Bookmarks

Deploy

Input Parameters

ΔH (Enthalpy)

50.00

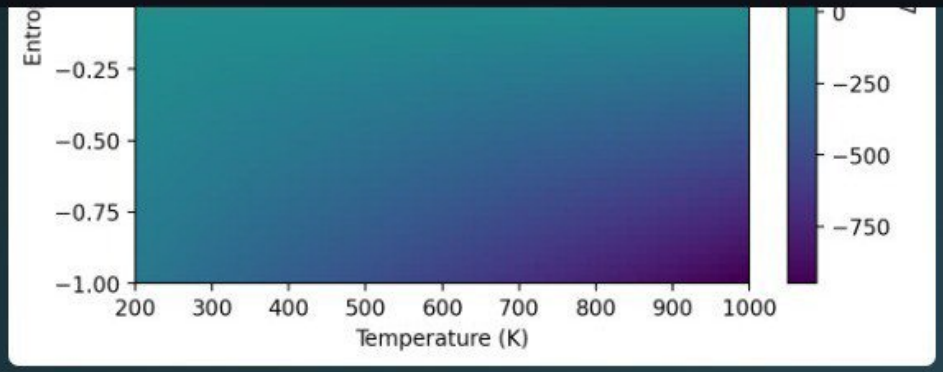
ΔS (Entropy)

0.20

Temperature (K)

300

Run Analysis



Interpretation

- Negative ΔG → Spontaneous region
- Positive ΔG → Non-spontaneous region
- The boundary between them represents equilibrium